Possible Interview questions

- Why would we choose to use a GLM instead of a classical OLS linear regression?
    - What are some of the main differences between a classical linear regression and a GLM?
    - Why would we always choose a N OLS instead of N GLM?
    - Why choose a N GLM
- What do we use GLMs for? (from a very basic perspective)
- What are the main components of a GLM? [**EXTRA:** Can you give a small explanation of each?]
- Why is it important for us to determine whether or not a distribution is from the 1-parameter exponential family?
- How do you determine whether or not a distribution belongs to the exponential family?
    - What does the mean-variance relationship in a GLM refer to?
    - How would you find out if that M-V relationship is violated?
- How do we decide on a random component for our GLM?
    - What is the gamma GLM used to model?
    - If I have continuous response data that seems to be positively skewed, which GLM options might be appropriate, and why?
- How do you find the canonical link function for a GLM? [Not the best answer – it's the same functional form as the natural parameter; better answer – 1st derivative of the log partition function, set equal to mu and invert.]
- What mathematical properties do we want from a link function? [continuous, monotonic, correct mapping]
    - Suppose I give you a completely new one-parameter distribution that I want you to use for the random component, and I give you a linear predictor. I want you to build and estimate a GLM using this random component. Can you guide me through the processes you would follow first to BUILD, then to ESTIMATE? [… you find it's in canonical form] [Get in exponential form, if canonical, get derivative of log partition function, set equal to mean, invert for link function. Find estimating equations.
- What does it mean when the natural parameter is canonical? [The sufficient statistic is y_i)
- Is there any reason to use non-canonical link functions?
- **EXTRA:** Why use canonical links?
- **EXTRA:** Advantages and disadvantages
- What is the score equation for a GLM?
- How does one determine the GLM MLE for the linear predictor's parameters?
    - What is the difference between the Newton-Raphson method and the Fisher Scoring method?
    - What do we use the Taylor series expansion for in GLM estimation?
    - What does the inverse of the Fisher Information give us?

- How does one interpret the slope parameter estimates for:
    - An identity link
    - A log link (parameter 0.2)
    - A logit link
    - An inverse link
    - A probit link
- Challenger data example – what can you tell me from the output?
- How can we **test** goodness of fit for a single GLM?
- What can we use the Wald test to do?
- How can we **test** multiple parameters simultaneously in a GLM?
    - What can you do with a Wald test in a GLM? [parameter tests, joint parameter tests, goodness of fit]
    - What is the advantage of the Wald over the LR?
    - What does it mean if the residual deviance of a model is very small?
- What are the differences between the Wald, deviance tests and LR tests?
- What does the dispersion parameter in a GLM signify? How is it related to the variance?
    - What is the problem with GLM dispersion?
- What are the DHARMa residuals, and how do we use them?

- What is a dose-response model?
    - How do you interpret the coefficients of a probit link?
- How do we apply a GLM to model the data in a contingency table?
    - What is the main purpose of loglinear models for contingency tables
    - How do we interpret the parameters for loglinear models?
    - What is an offset in terms of loglinear models? [With count data, it's the log of exposure period, coefficient of 1][We use it to model count data, where the counts are determined as a relationship with the exposure period] [For contingency tables, it represents the fixed column or row margins or total]
    - How do we test the assumption of independence?
    - In a loglinear model, how do we perform tests of independence between variables (using GLMs)?